

BERLIN SCHONEFELD, Germany

WMO: 103850

Lat: 52.3808N

Lon: 13.5306E

Elev: 50

StdP: 100.72

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-8.2	-15.0	1.0	-9.3	-12.1	1.3	-7.2	13.4	6.4	12.3	6.3	3.4	60	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.2	19.4	29.2	18.7	27.2	18.3	20.9	27.5	20.1	26.3	19.3	25.2	4.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.9	22.6	18.1	13.1	22.0	17.1	12.3	21.4	60.4	27.3	57.5	26.5	54.9	25.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.4	8.3	-13.9	34.9	4.5	2.1	-17.1	36.4	-19.7	37.7	-22.2	38.9	-25.4	40.4		
(5)			-14.1	22.8	4.5	0.8	-17.4	23.4	-20.0	23.8	-22.5	24.3	-25.7	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	0.3	1.1	4.8	10.4	14.1	17.9	20.0	19.1	15.4	10.2	5.7	2.5		
	DBStd	7.90	5.03	4.17	4.30	4.02	4.07	3.59	3.36	3.21	3.17	3.53	3.66	4.34		
	HDD10.0	1178	301	249	166	44	8	0	0	0	1	42	136	232		
	HDD18.3	3163	559	482	420	239	142	49	21	28	100	252	381	491		
	CDD10.0	1242	0	0	4	55	136	237	312	281	162	49	6	1		
	CDD18.3	184	0	0	0	1	11	37	74	51	11	0	0	0		
	CDH23.3	1758	0	0	0	13	131	377	693	448	95	2	0	0		
	CDH26.7	558	0	0	0	1	32	118	239	153	16	0	0	0		
(14) Wind	WSAvg	4.1	4.8	4.4	4.4	4.0	3.9	3.8	3.7	3.5	3.7	3.8	4.1	4.6		
(15) Precipitation	PrecAvg	586	47	38	40	31	56	59	76	57	50	40	45	48		
	PrecMax	785	92	84	97	78	122	166	207	122	135	100	99	103		
	PrecMin	414	3	6	9	3	5	8	21	23	15	8	1	13		
	PrecStd	95	22	19	23	16	31	33	41	26	28	24	21	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.1	19.0	25.0	30.1	32.8	34.8	34.0	29.0	22.9	15.9	12.8		
		MCWB	9.6	7.3	11.4	15.0	18.3	19.9	19.7	19.9	18.7	16.4	13.6	10.5		
	2%	DB	10.0	10.1	15.9	22.2	26.9	29.9	31.2	30.9	26.0	19.8	13.9	10.9		
		MCWB	8.5	7.7	9.9	13.4	17.3	19.0	19.1	19.9	17.5	14.8	11.8	9.3		
	5%	DB	8.2	8.8	13.2	20.2	24.1	27.8	29.2	27.9	23.9	17.9	12.1	9.8		
		MCWB	7.0	6.6	8.7	12.7	15.7	18.6	18.9	18.7	16.6	14.0	10.1	8.3		
	10%	DB	7.0	7.0	11.1	18.1	22.1	25.1	27.1	25.2	21.2	15.9	10.9	8.1		
		MCWB	6.0	5.2	7.8	11.7	14.4	17.1	18.4	17.6	15.6	13.1	9.2	7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	9.5	12.5	15.9	19.7	21.6	21.7	21.7	19.9	17.1	13.7	11.0		
		MCDB	11.5	10.1	17.2	22.7	26.4	28.9	28.6	29.1	26.6	20.9	15.6	12.5		
	2%	WB	8.6	7.7	10.9	14.4	18.3	20.3	20.9	20.7	18.3	15.6	12.0	9.4		
		MCDB	9.8	9.5	14.6	20.2	24.9	26.6	27.6	27.1	23.7	18.9	13.6	10.3		
	5%	WB	7.1	6.4	9.6	13.3	16.9	19.2	20.1	19.8	17.3	14.3	10.6	8.2		
		MCDB	8.0	8.2	12.4	19.1	22.2	25.3	26.3	25.6	22.1	17.0	12.0	9.3		
	10%	WB	5.8	5.0	8.2	12.2	15.7	18.1	19.3	18.8	16.3	13.2	9.5	6.9		
		MCDB	6.7	6.7	10.8	17.7	20.5	23.5	25.5	23.9	20.6	15.5	10.6	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	6.0	8.1	10.6	10.6	10.5	10.1	10.2	9.6	7.7	5.4	4.6		
		MCDBR	5.4	8.6	12.4	13.8	14.1	14.1	14.0	14.0	13.2	10.4	6.9	5.8		
	5% WB	MCWBR	4.6	6.3	7.5	7.1	6.1	5.6	4.9	5.4	6.0	6.2	5.1	4.7		
		MCWBR	5.1	7.8	10.8	12.6	12.6	12.3	11.6	11.3	11.2	9.3	6.4	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.353	0.394	0.419	0.416	0.413	0.434	0.428	0.389	0.376	0.351	0.320		
		taud	2.310	2.286	2.193	2.198	2.250	2.283	2.236	2.268	2.351	2.347	2.352	2.326		
	Ebn at Noon		653	745	782	812	836	841	813	797	785	710	618	592		
		Edh at Noon	64	87	115	129	129	126	130	119	98	81	62	54		
(44) All-Sky Solar Radiation	RadAvg		0.64	1.34	2.36	3.94	4.87	5.18	4.86	4.18	3.05	1.70	0.78	0.48		
	RadStd		0.07	0.23	0.27	0.51	0.58	0.41	0.64	0.34	0.35	0.20	0.10	0.05		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (39 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page